



Enhancing a Computational Pathology Model Based on Detailed Performance Assessment

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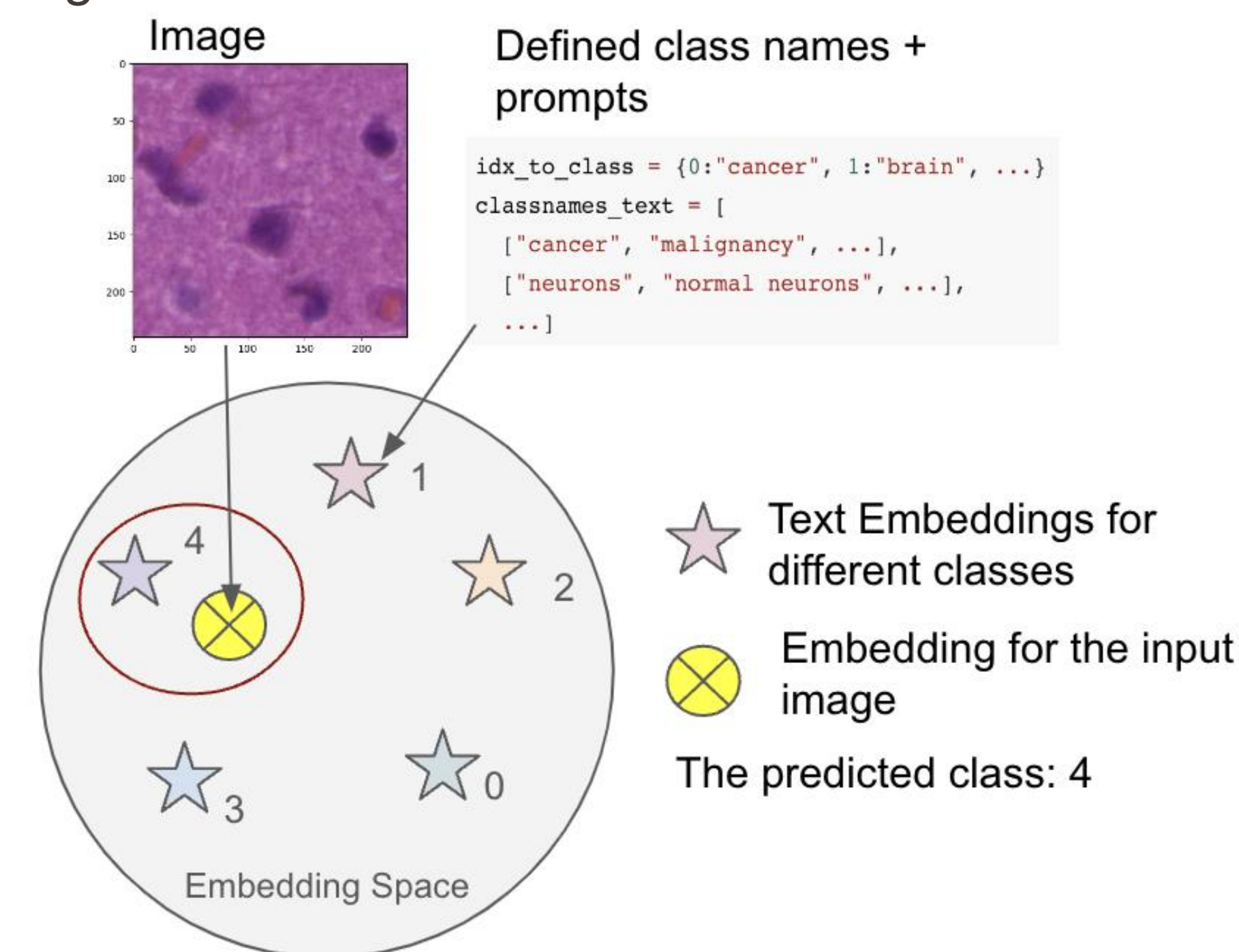
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Background

- Examination of tissues by pathologists remains the golden standard of cancer pathology¹.
- Machine learning models play an increasingly important role in tasks such as cancer subtyping².
- Training a digital pathology machine learning model for a specific task is time- and labor-consuming; therefore, a foundation model is needed¹.
- CONCH is a foundation model trained with 1.17 million image-caption pairs to generalize across diverse histopathology tasks¹.

Model Overview

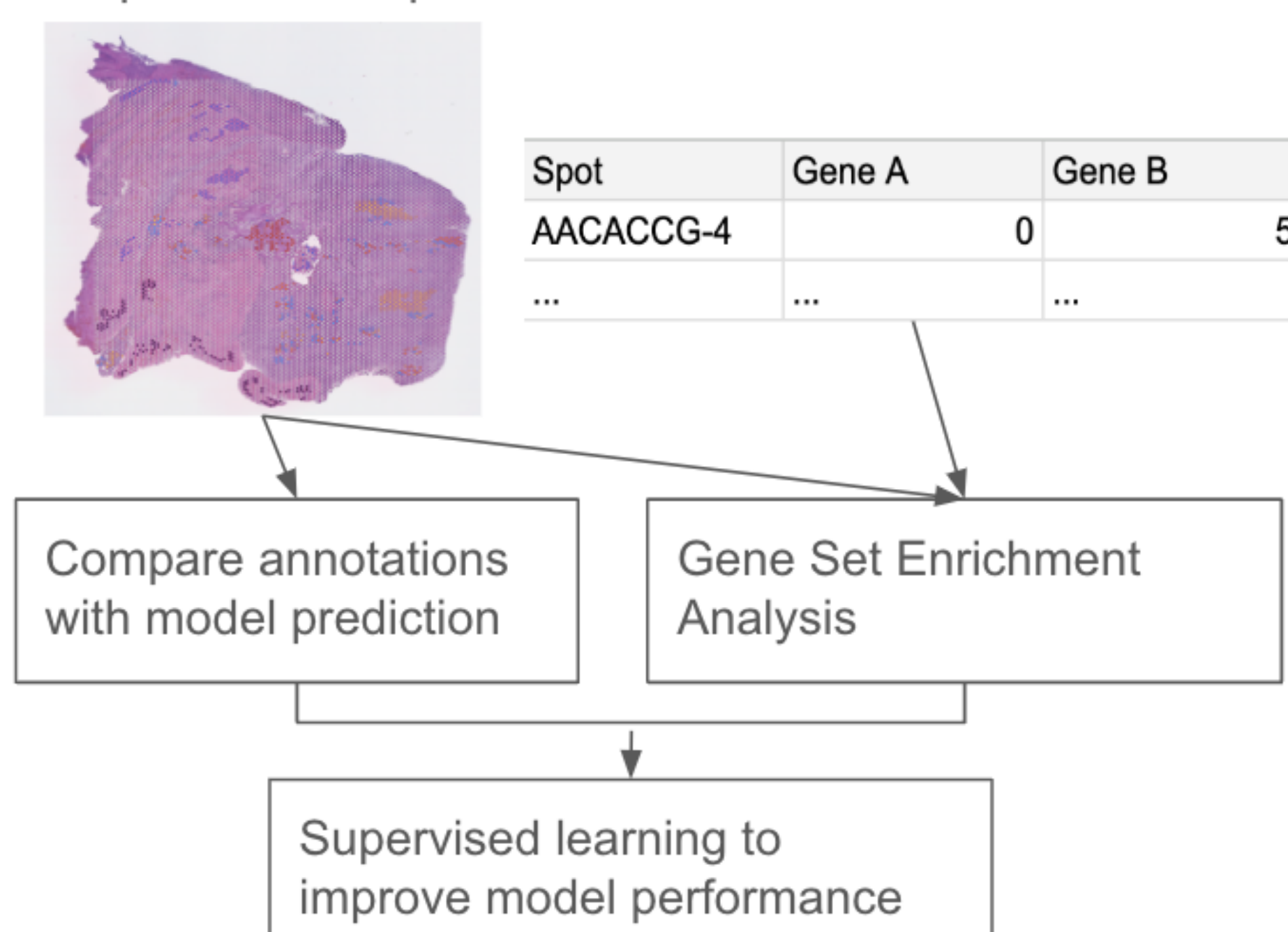
Fig. 1 Overview of CONCH Zero-Shot Classification



Methods Overview

Fig. 2 Overview of research methods

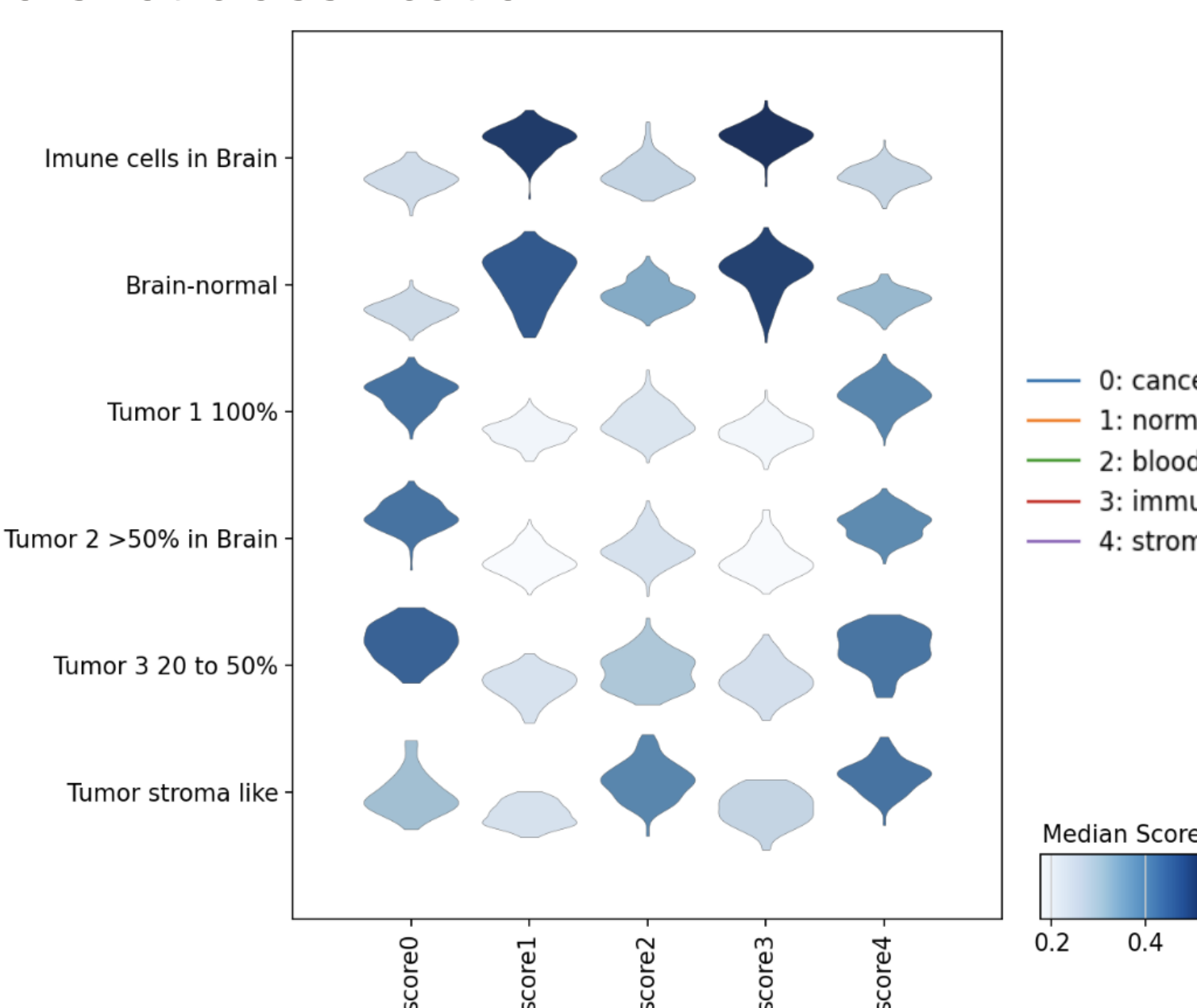
Available Data: Image with some parts manually annotated + Spatial transcriptomics data



- CONCH zero-shot prediction results were compared with the pathologist's manual annotations.
- After integrating spatial transcriptomics data, gene set enrichment analysis was performed among different CONCH predicted classes.
- Logistic regression was applied to improve CONCH zero-shot classification performance.

Comparison with Annotations

Fig. 3 Distribution of CONCH zero-shot probability scores for each class type in annotations using CONCH zero-shot classification



- High performance in identifying cancer cells.
- Less accuracy in distinguishing immune cells from normal brain cells.

Gene Set Enrichment Analysis

Fig. 4 Method of gene set enrichment analysis

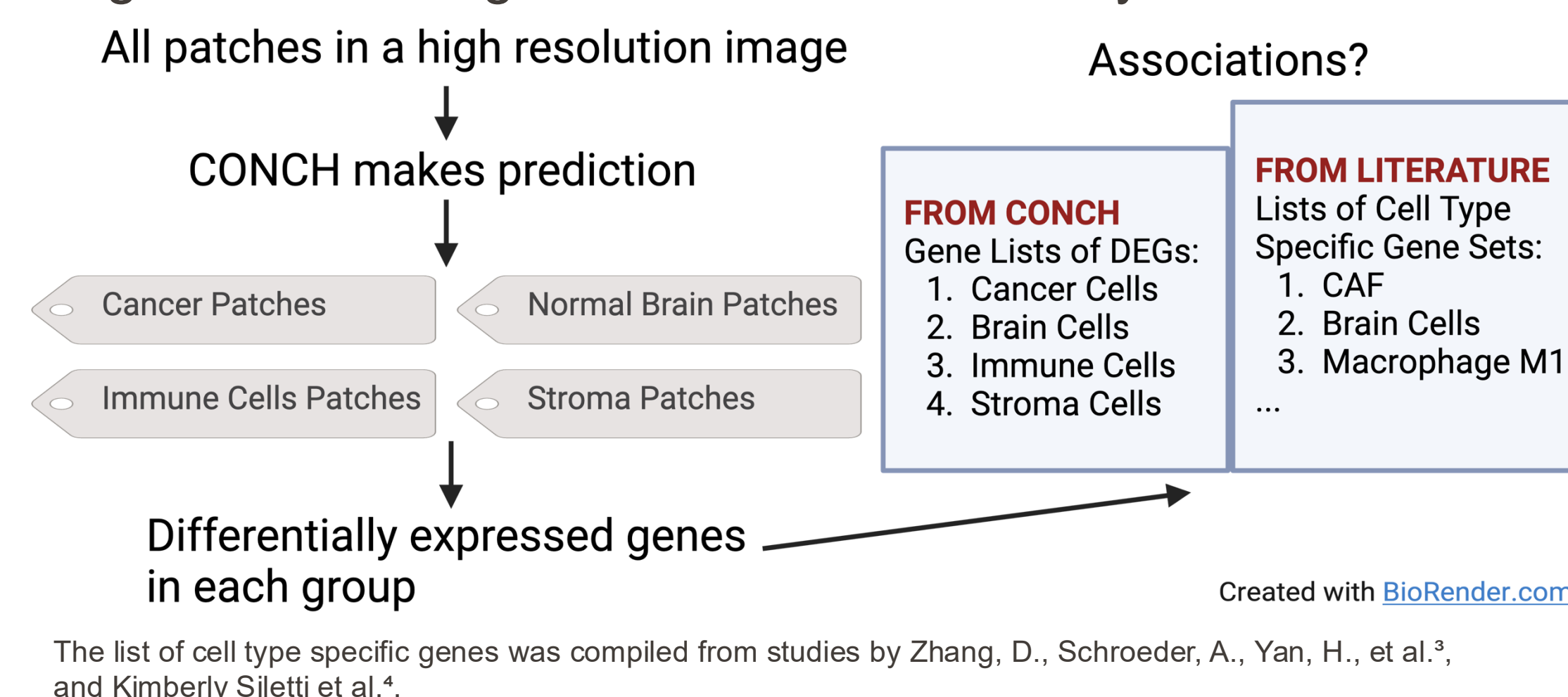
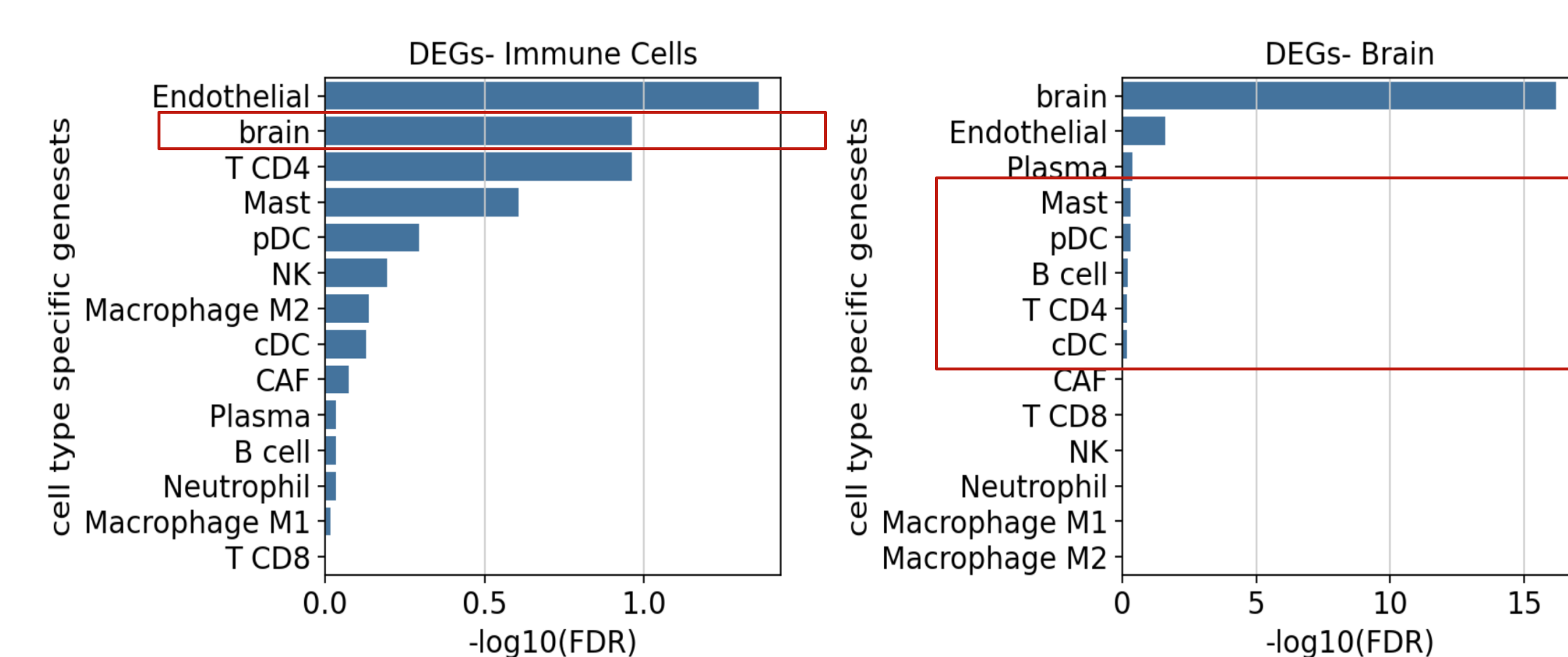


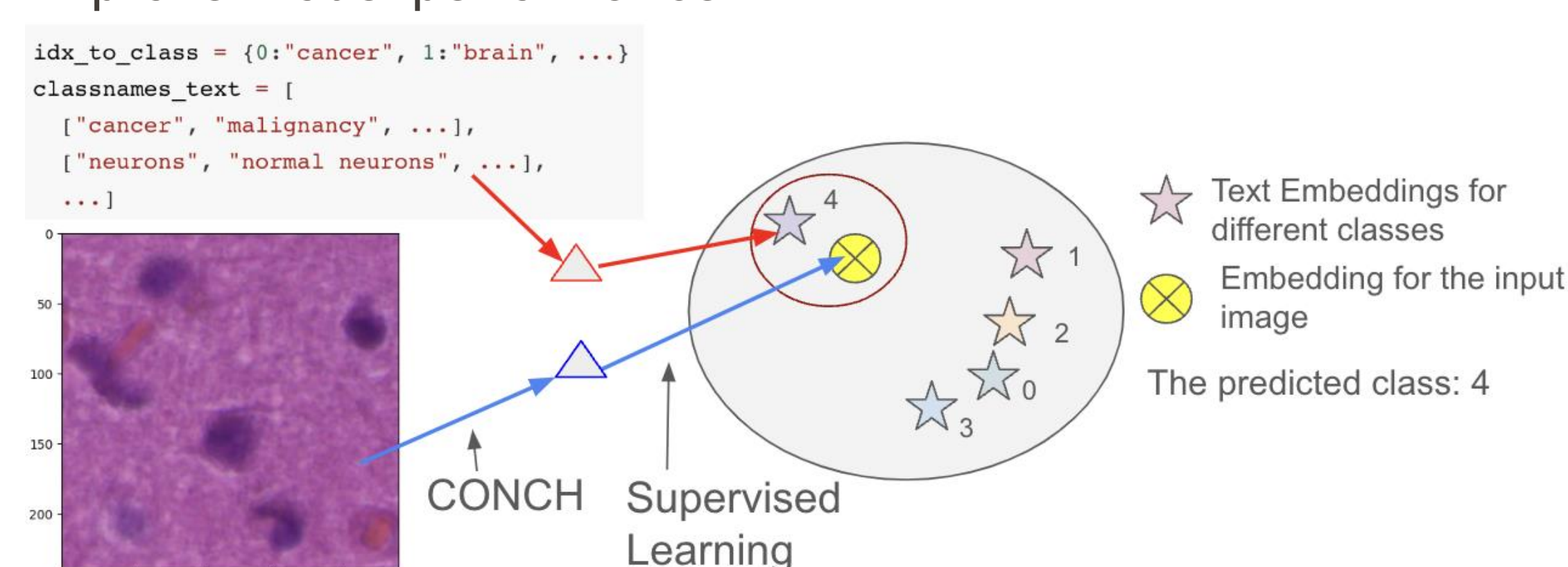
Fig. 5 Results of gene set enrichment analysis using CONCH zero-shot classification



- Validated: CONCH is less effective in differentiating between immune cells and brain cells.

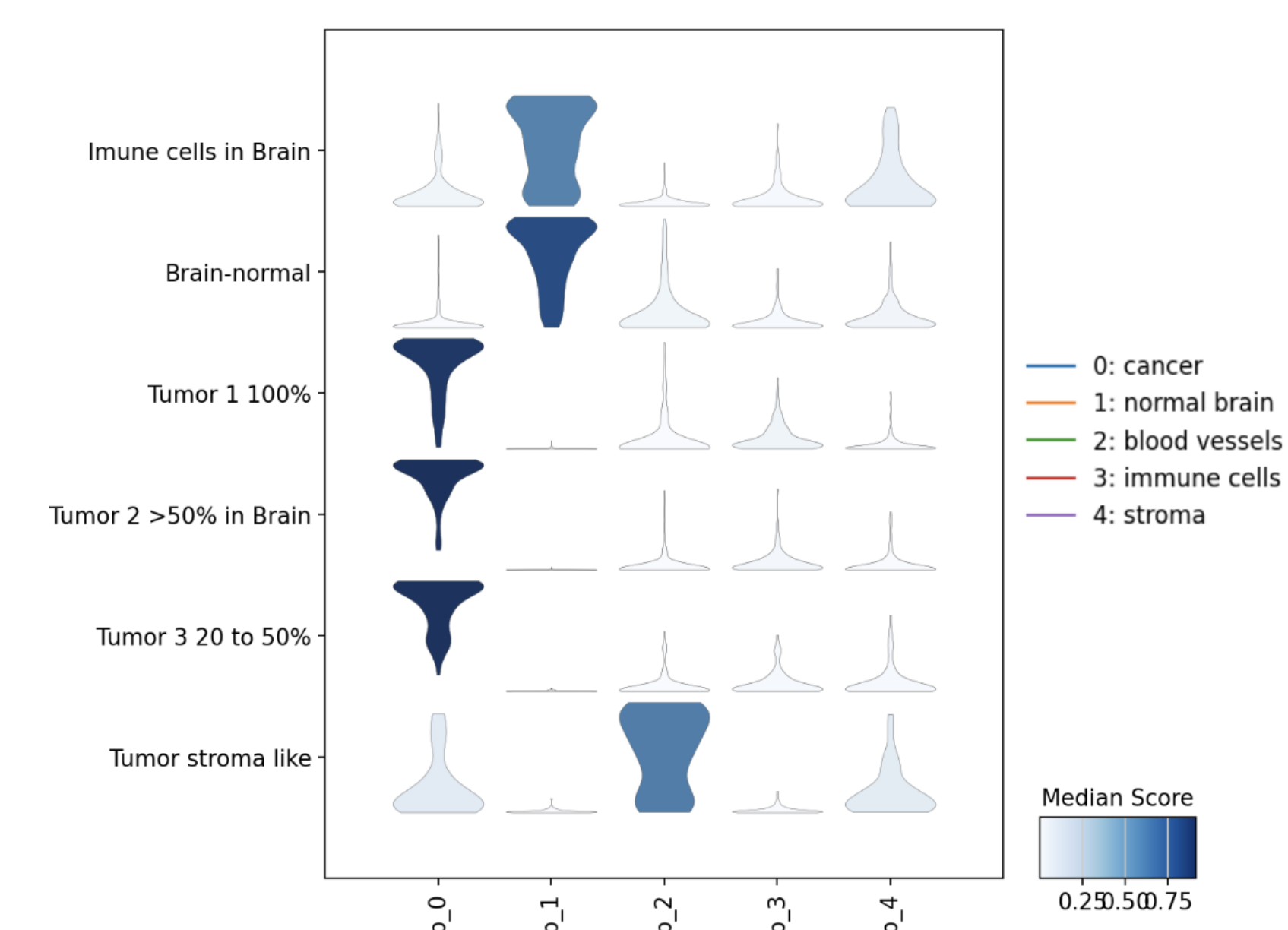
Improving Model Performance

Fig. 6 Method of using task-specific supervised learning to improve model performance



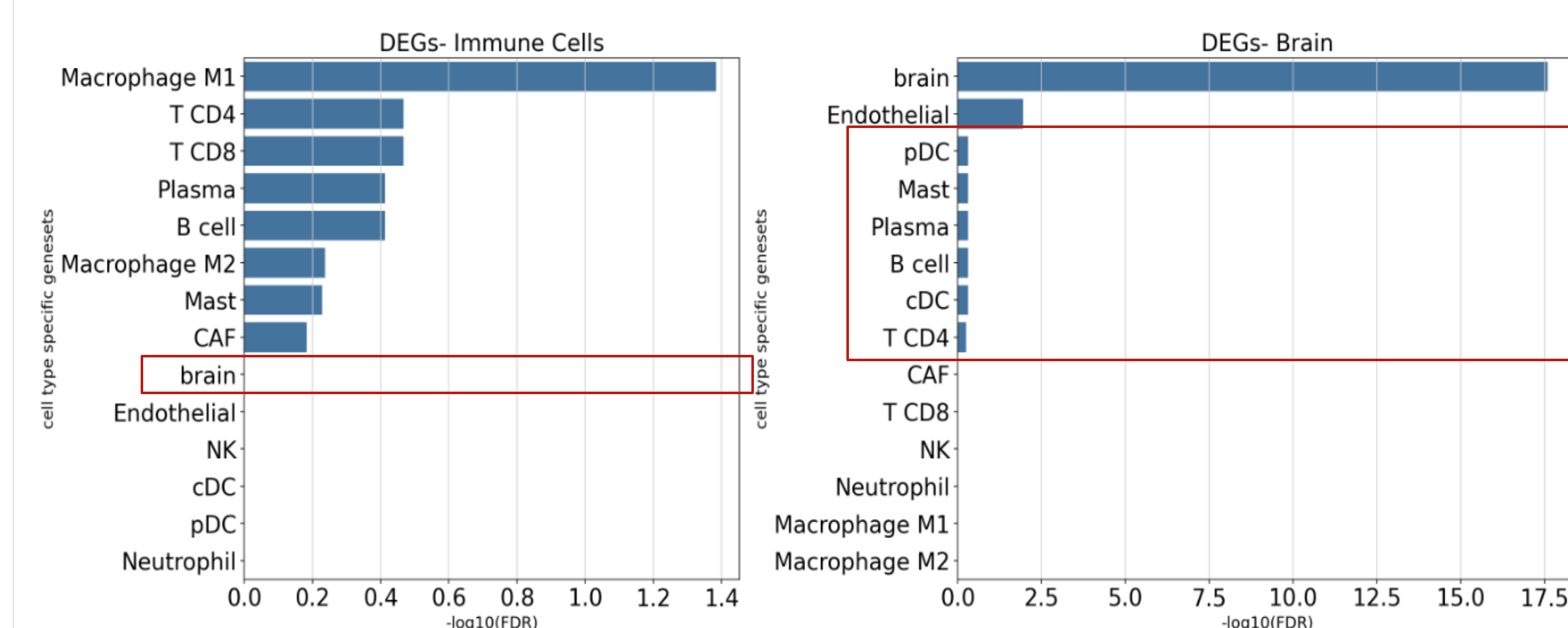
Improving Model Performance (Cont'd)

Fig. 7 Distribution of CONCH classification probability scores for each class type in annotations using the enhanced model



- Better accuracy in identifying tumors and normal brain cells.
- Drawback: immune cell patches falsely identified as normal brain cells.

Fig. 8 Results of gene set enrichment analysis using CONCH model with logistic regression



- Brain-cell-specific genes no longer associated with differentially expressed genes (DEGs) identified among patches predicted to be immune cells.
- For immune cell spots: the new model has a high false negative rate.
 - Possible reason: mixtures of immune cells and normal brain cells in the images in the training dataset.

Conclusions and Next Steps

- CONCH zero-shot classification shows high performance in identifying cancer cells, though it performs less effectively in distinguishing between brain cells and immune cells.
- Fine-tuning with CONCH-extracted embeddings enhances the model's overall performance, though it leads to a higher false negative rate when making predictions for images of immune cells in the brain.
- Next steps:
 1. A better dataset with less mixture of different cell types.
 2. Use annotations from multiple pathologists.
 3. Integrate high-resolution stacked images.

Reference

1. Lu, M.Y., Chen, B., Williamson, D.F.K. et al. A visual-language foundation model for computational pathology. Nat Med 30, 863–874 (2024).
2. Coudray, N. et al. Classification and mutation prediction from non-small cell lung cancer histopathology images using deep learning. Nat. Med. 24, 1559–1567 (2018).
3. Zhang, D., Schroeder, A., Yan, H. et al. Inferring super-resolution tissue architecture by integrating spatial transcriptomics with histology. Nat Biotechnology (2024).
4. Kimberly Siletti et al. Transcriptomic diversity of cell types across the adult human brain. Science 382, eadd7046 (2023). DOI:10.1126/science.add7046